ICP 06. Bedside Management and Postpartum Follow-Up

Companion reading handout for simultaneous listening.
Audio course on intrahepatic cholestasis of pregnancy. Episode six. Bedside management and postpartum follow-up.

Source Base

This episode is based on the Israeli ICP position paper, Gabbe's hepatic disease and fetal surveillance chapters, Ovadia two thousand nineteen, and the SMFM consult.

This final episode turns the course into a bedside algorithm.

The algorithm is not a checklist to memorize. It is a sequence of clinical reasoning.

Diagnose the pattern. Stratify by bile acids. Treat symptoms. Monitor honestly. Time delivery by peak risk. Confirm resolution.

Management Sequence

Let us walk through that sequence.

Step one. Suspect ICP.

The patient is usually in the late second trimester or third trimester. She has pruritus without a primary rash, often palms and soles, often worse at night. Ask about prior ICP, family history, multifetal pregnancy, IVF, hepatobiliary disease, gallstones, medications, and previous liver disease.

At the same time, check for clues against uncomplicated ICP. Blood pressure. Headache. Visual symptoms. Epigastric pain. Right upper quadrant pain. Fever. Vomiting. Jaundice. Rash morphology. Medication exposure. These are not administrative questions. They decide whether this is ICP or a mimic.

Step two. Send the right tests.

The core tests are total bile acids and AST and ALT. Add bilirubin. Interpret alkaline phosphatase cautiously because it normally rises in pregnancy. If the case is severe, jaundiced, painful, hypertensive, systemically ill, or atypical, add platelets, creatinine, urine protein assessment, coagulation studies, viral testing, and imaging as clinically indicated.

If bile acids are above ten micromoles per liter in a symptomatic patient, the diagnosis is supported. If they are normal but the symptoms are classic and persistent, repeat bile acids and transaminases. Do not close the case too early.

Step three. Classify by peak bile acids.

This is where residents must be disciplined. Do not classify by itch severity. Do not classify by ALT. Do not classify by how anxious the room feels.

Use peak total bile acids.

Below forty is the lower-risk group. Forty to ninety-nine changes the tempo and moves delivery earlier. One hundred or higher changes the risk conversation.

Remember the Ovadia numbers. In singleton pregnancies, stillbirth prevalence was about zero point one three percent below forty, zero point two eight percent from forty to ninety-nine, and three point four four percent at one hundred or higher.

Step four. Treat the mother.

Treatment

Start ursodeoxycholic acid, or UDCA, for maternal symptoms. A practical starting approach is ten to fifteen milligrams per kilogram per day in divided doses. The Israeli paper gives three hundred milligrams three to four times daily. If needed, titrate up to twenty-one milligrams per kilogram per day.

Tell the patient what to expect. Itch may improve in one to two weeks. Labs may improve in three to four weeks. Also tell her what not to expect. UDCA is not proven to prevent stillbirth, and it does not replace delivery planning.

If symptoms are refractory, reassess the diagnosis, optimize UDCA, and involve maternal-fetal medicine. Consider rifampin or rifampicin as an add-on in selected refractory cases. Consider cholestyramine when appropriate, but remember gastrointestinal effects and vitamin K absorption. Consider S-adenosyl methionine as a less central adjunct. Use antihistamines for sleep, not as disease treatment. Check coagulation or vitamin K status when jaundice, steatorrhea, cholestyramine use, or bleeding risk makes it relevant.

Step five. Monitor honestly.

The Israeli position paper says continued maternal-fetal medicine follow-up is recommended, including fetal heart monitoring by gestational age and ultrasound follow-up. It also states that no monitoring strategy or hospitalization has been proven to prevent sudden fetal death, and that monitoring frequency is left to clinical discretion.

Maternal monitoring also matters. The Israeli paper notes an increased incidence of preeclampsia in women with cholestasis, often beginning two to four weeks after the ICP diagnosis. So follow-up is not only fetal monitoring and bile acids. It is also blood pressure, symptoms, platelets and liver tests when clinically indicated, and vigilance for a second disease process.

SMFM suggests beginning antenatal fetal surveillance when the gestational age is one at which an abnormal result would lead to delivery, or at diagnosis if diagnosis occurs later.

Gabbe's fetal surveillance chapter lists weekly modified BPP from thirty-four weeks, while older hepatic-chapter language mentions twice-weekly NST as reasonable. But the most important teaching is not the exact frequency. The most important teaching is humility.

ICP stillbirth may be sudden. It may occur after reassuring testing. It is not usually preceded by growth restriction, oligohydramnios, or classic chronic placental histology.

So use surveillance. But do not let surveillance become false reassurance. A normal NST does not erase bile acids of one hundred and twenty.

Step six. Plan delivery.

Under Israeli guidance, use the highest measured bile-acid level.

If peak bile acids are below forty, deliver at thirty-seven plus zero to thirty-eight plus six.

If peak bile acids are forty to ninety-nine, deliver at thirty-six plus zero to thirty-seven plus zero.

If peak bile acids are above one hundred, deliver at thirty-four plus zero to thirty-six plus zero.

Delivery at thirty-four to thirty-six weeks can also be considered with elevated bile acids plus pruritus unresponsive to treatment, prior fetal death at an earlier gestational age in ICP, or departmental discretion.

Under SMFM, bile acids below one hundred are delivered between thirty-six plus zero and thirty-nine plus zero, with below forty toward the later end and forty to ninety-nine earlier. Bile acids at or above one hundred are offered delivery at thirty-six plus zero. Delivery between thirty-four and thirty-six weeks is reserved for one-hundred-or-higher disease with special high-risk features, such as excruciating refractory pruritus, prior early ICP stillbirth, or worsening hepatic disease.

Before delivery before thirty-seven weeks, give antenatal corticosteroids if not already given and if appropriate. Counsel explicitly about prematurity. The decision is not bile acids alone. It is bile acids plus gestational age, symptoms, prior history, comorbidities, fetal status, local policy, and patient values.

Step seven. Manage labor with awareness of fetal risk.

ICP is associated with meconium-stained fluid and fetal heart rate abnormalities. SMFM recommends continuous fetal monitoring during labor. This does not mean every patient needs cesarean delivery. It means intrapartum fetal status matters, and the team should not treat the labor as low risk.

Step eight. Follow postpartum resolution.

Delivery usually produces rapid improvement. Gabbe states that delivery rapidly and completely relieves cholestasis in typical cases. But do not forget the postpartum check.

If pruritus persists for four to six weeks after delivery, or if liver tests remain abnormal, SMFM recommends repeat biochemical testing and referral to a liver specialist if abnormalities persist. That is because pregnancy may have revealed underlying hepatobiliary disease rather than creating the entire problem.

Counsel about recurrence. The Israeli position paper reports recurrence up to ninety percent, often earlier in later pregnancies. Gabbe reports recurrence around two thirds. Either way, recurrence risk is high. In a future pregnancy, early pruritus deserves prompt bile acids and transaminases.

Also counsel about hepatobiliary morbidity. ICP is pregnancy-limited in the usual case, but it is also a marker that the hepatobiliary system may be vulnerable.

Clinical Cases

Now let us finish the course with five clinical cases.

Case one. Thirty weeks, classic itching, bile acids eight.

Not confirmed. Repeat bile acids and AST and ALT if symptoms persist. Consider empiric symptom treatment if severe, but understand that starting UDCA before labs return can make biochemical confirmation harder.

Case two. Thirty-two weeks, bile acids twenty-six.

Confirmed ICP. Treat symptoms, repeat labs, follow, and do not deliver yet solely because ICP exists.

Case three. Thirty-five weeks, bile acids fifty-eight.

This is the forty to ninety-nine group. Under local Israeli guidance, plan delivery thirty-six plus zero to thirty-seven plus zero. Give steroids if delivering before thirty-seven and not already administered.

Case four. Thirty-five weeks, bile acids one hundred and thirty, severe pruritus despite medication.

High-risk disease. Local Israeli guidance supports delivery from thirty-four to thirty-six weeks. SMFM supports thirty-six plus zero generally, with thirty-four to thirty-six considered when severe refractory symptoms or other high-risk features are present. This is an MFM-level conversation.

Case five. Postpartum, still itching at six weeks, ALT still elevated.

Do not call this lingering ICP and move on. Repeat labs and refer for hepatobiliary evaluation.

Course Summary

Now final synthesis of the whole course.

First, ICP is a pattern: late pregnancy itching, usually no primary rash, and elevated bile acids.

Second, the differential diagnosis matters. ICP is uncomfortable for the mother, but its mimics can be maternal emergencies.

Third, fetal risk is bile-acid driven. Peak total bile acids are more important than ALT or itch severity.

Fourth, the thresholds are ten, forty, and one hundred. Ten names it. Forty changes the tempo. One hundred changes the risk conversation.

Fifth, UDCA treats maternal symptoms and labs. It is not proven stillbirth prevention.

Sixth, surveillance is useful but imperfect. It should include maternal vigilance as well as fetal testing, because preeclampsia can appear after ICP diagnosis.

Seventh, delivery timing is the central fetal-risk intervention.

The final memory sentence is this.

The itch brings her to you. The bile acids stratify the fetus. The peak value times the delivery. And postpartum resolution confirms the diagnosis.

End of episode six.